Available online at www.sciencedirect.com
ScienceDirect
Proceedings of the Combustion Institute 31 (2007) 1625–1632
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

NO_x emission characteristics of confined jet nonpremixed flames

Susumu Noda *, Junya Inohae, Zaki S. Saldi

Department of Mechanical Engineering, Toyohashi University of Technology, Toyohashi 441-8580, Japan

Abstract

The flame characteristics of confined flames are investigated for propane and hydrogen jet nonpremixed flames in cylindrical furnaces. The flame characteristics can be strongly dominated by the combination of the burner and furnace geometries. In the present study, the effects of the inner diameter of the cylindrical furnace D_1, the turbulence at the flame boundary, and the global equivalence ratio ϕ are examined in terms of the emission of NO_x. The emission index of NO_x, $EINO_x$, decreases roughly with these parameters. The decrease in $EINO_x$ is thought to be related to the dilution of mixtures by the burned gas and the flame stretch. The dilution is attributable to vortices formed at the bottom of the furnace, and the flame stretch is attributable to the air velocity difference ΔU_a created by two air nozzles. The $EINO_x$ characteristics are evaluated by the furnace Reynolds number Re, c based on the inner diameter of the furnace reflecting the dilution and the strain rate $S \sim Da^{-1}$ based on the air velocity difference, where Da indicates the Damköhler number. The $EINO_x$ is well characterized by the factor $D_1 U_F \Delta U$, which is proportional to Re, cDa^{-1}. This result indicates that the confined nonpremixed flame is characterized by the furnace volume, not the flame volume, which is related to the reaction rate of the jet nonpremixed flame with coflowing air based on the flamelet model. This may imply that the furnace combustion should be modified from the flamelet concept owing to the dilution.
© 2006 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

Keywords: Confined turbulent nonpremixed flame; NO_x emission; Modeling of confined combustion

1. Introduction

The reduction of pollution generated by combustion is an urgent problem. NO_x reduction technologies, including a new high-temperature air combustion (HiTAC) technology [1], have been continuously explored. Scaling laws of NO_x emission for open jet nonpremixed flames have been examined extensively in order to determine a baseline for evaluating combustion facilities. Peters and Donnerhack [2] constructed a scaling law based on the flamelet model under the equilibrium assumption to make the total production rate of NO_x proportional to the flame volume. The emission index of NO_x denoted by $EINO_x$ is then given as

$$EINO_x = cL_F^3/(\rho_F U_F d_F^2), \tag{1}$$

where L_F is the flame length, ρ_F is the density of fuel, U_F is the mean velocity of fuel, and d_F is the inner diameter of the fuel nozzle. On the other hand, Chen and Driscoll [3] modified Eq. (1) to involve the effects of the wrinkled flame area and the flame stretch, which are related to the Reynolds

* Corresponding author. Fax: +81 532 44 6661.
E-mail address: noda@mech.tut.ac.jp (S. Noda).

1540-7489/$ - see front matter © 2006 The Combustion Institute. Published by Elsevier Inc. All rights reserved.
doi:10.1016/j.proci.2006.07.088

number Re and the Damköhler number Da, respectively, as

$$\mathrm{EINO}_x/(L_\mathrm{F}^3/(U_\mathrm{F}d_\mathrm{F}^2)) = cRe^{m-1/2}Da^n. \quad (2)$$

The denominator on the left-hand side of Eq. (2) denotes the residence time of fuel passing through the flame. Thus, Eq. (2) also indicates the dependence of Re and Da on the production rate of NO_x. From an experiment involving hydrogen jet nonpremixed flames, they concluded that $m = 1/2$ and $n = -1/2$ and demonstrated a linear logarithmic relationship between $\mathrm{EINO}_x/(L_\mathrm{f}^3/d_\mathrm{F}^2U_\mathrm{F})$ and $U_\mathrm{F}/d_\mathrm{F}$ ($\sim Da^{-1}$). The exponents in Eq. (2) however may appear differently for other fuels. Moreover, Turns [4], Kim et al. [5], and Santos and Costa [6] discussed the effect of the radiation based on the deviation from Eq. (2). These investigations have not only generated scaling laws but have also provided insight and understanding with regard to flame structures. However, in practical use, combustion technologies related to NO_x reduction may become more important in confined combustion systems, such as the above-mentioned HiTAC. For confined combustion, the dilution of fuel and air by the burned gas becomes important, although this effect was not considered in the above studies. This effect should be considered in the formulation of confined combustion. However, few NO_x scaling studies on confined combustion systems have been reported [7]. The present study examines the NO_x emission characteristics of propane and hydrogen jet nonpremixed flames confined in cylindrical furnaces and attempts to identify the NO_x emission through primarily experimental means. The flame characteristics can be strongly dominated by the combination of the burner and the furnace geometries. Therefore, the effects of the inner diameter of the cylindrical furnace D_1, the turbulence at the flame boundary, and the global equivalence ratio ϕ are examined in terms of the emission of NO_x.

2. Experimental method

Figure 1 shows a schematic diagram of a cylindrical furnace with an expanded burner. The experimental apparatus was installed vertically. The side wall of furnace is made of a cylindrical Pyrex tube, which can easily be replaced for different wall sizes: inner diameters (i.d.) of $D_1 = 95$ mm (rim thickness: 2.5 mm), 142 mm (3.5 mm), and 182 mm (4.0 mm) in order to view the flame. Each glass tube was installed between lower and upper flanges. The height of the furnace is 840 mm. The exit is contracted to 38 mm to ensure a uniform concentration field for the NO_x measurement. The burner consists of a fuel nozzle with two air coaxial nozzles. The fuel nozzle is made of a stainless pipe of 2 mm i.d. with a rim thickness of 0.2 mm. A pilot nozzle of 3.19 mm i.d. with a rim thickness of 0.3 mm is installed and is used for propane flames, but not for hydrogen flames. The air nozzles consist of an inner higher-velocity nozzle and an outer lower-velocity nozzle, which are also made of stainless steel pipes. The higher-velocity air nozzle is of 12 mm i.d. with a rim thickness of 1 mm, and the lower-velocity nozzle is of 30 mm i.d. with a rim thickness of 1 mm. The air nozzles allow the change in turbulence at the flame boundary due

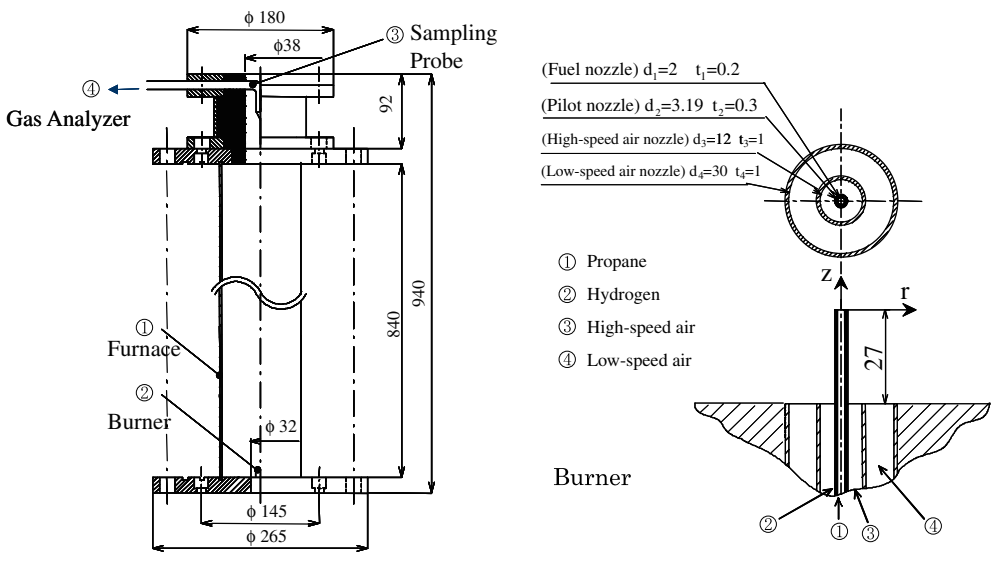

Fig. 1. Furnace with replaceable cylindrical Pyrex walls and expanded burner composed of a fuel nozzle surrounded by high-velocity and low-velocity air nozzles.

to the air velocity difference. The tip of fuel nozzle is located 27 mm downstream from the furnace base, where the air nozzle exits are located. This configuration ensures the interaction between the flame and the turbulence generated by the airflows.

Photographs of flames were taken directly by a digital camera (OLYMPUS, C-5050ZOOM) with an exposure of 1/5 s and F2.3. Concentration measurements were performed at the exit of the furnace using an NO_x–O_2 gas analyzer (SHIMAZU, NOA-7000) for NO_x (NO and NO_2) and an infrared gas analyzer (SHIMAZU, CGT-7000) for CO, CO_2, O_2, and CH_4. A sampling probe of 2 mm i.d. was cooled by water in order to freeze the reaction and withstand the high temperatures. The sampling line between the probe and the analyzer however was heated to over 100 °C in order to avoid condensation of H_2O, which dissolves NO_x. Moreover, a preprocessing unit, referred to as a gas sampling unit (SHIMAZU, CFP-8000), was used to remove particles such as water and soot. Every measurement was implemented twice to ensure the repeatability of data. In the present study, NO_x data is discussed only for the flame condition in which the CO concentration is lower than 300 ppm.

Flame characteristics were examined with respect to the inner diameter of the furnace D_1, the global equivalence ratio ϕ, and the air velocity difference ΔU_a. The experimental conditions are given in Tables 1 and 2. The fuels used in the present study are propane (C_3H_8 98.2 vol%, C_4H_{10} 1.0 vol%, and C_2H_4 0.8 vol%) and hydrogen (H_2 99.99 vol%). The fuel of the pilot flame is hydrogen, and the oxidizer is air. The total airflow rate was fixed at 120 l/min. Thus, the global equivalence ratio ϕ is determined by the fuel flow rate. A larger ϕ indicates a higher-velocity of the fuel flow. For propane flames, the flow rate of the hydrogen as the fuel of the pilot flame was determined on the basis of the ratio of the hydrogen flow rate Q_{H_2} to the total fuel flow rate Q_f being $Q_{H_2}/Q_f = 0.3$. The air velocity difference ΔU_a is based on both the mean velocities of the higher-velocity and lower-velocity air flows and reflects the turbulence intensity at the flame boundary. The Reynolds number Re in Table 1 is based on the fuel mean velocity and the inner diameter of the fuel nozzle.

For the confined combustion, the dilution of fuel and air by the burned gas becomes important as mentioned above. This effect should be considered in the formulation of the furnace combustion. We tried to characterize the NO_x emission using various parameters, including the parameters explained in Section 1, and two parameters were finally determined to be appropriate: the Reynolds number Re, $c = D_1 U_F/\nu$ based on the furnace inner diameter and the Damköhler number Da based on the air velocity difference, which is inversely proportional to the stretch rate $S = \Delta U_a/\Delta L$, where ΔL is the distance between the two air nozzles. The Reynolds number Re, c denotes the turbulence in the furnace and the strength of circulation of the vortex formed at the bottom of the furnace. Consequently, Re, c reflects the level of dilution by the burned gas. The stretch rate S indicates the flame stretch and the mixing strength through the enhanced turbulence mixing.

3. Numerical support

A calculation was performed in order to support the discussion, because velocity fields were not measured experimentally. The calculation,

Table 1
Experimental conditions for fuels

Global equivalence ratio ϕ		0.4	0.6	0.8	0.9
(a) Propane + pilot flame (H_2)					
Propane	$U_{C_3H_8}$ [m/s]	10.4	15.5	20.7	23.3
Pilot flame (H_2)	U_{H_2} [m/s]	4.0	6.0	8.0	9.1
Reynolds Number	Re	4500	6800	9100	10200
(b) Hydrogen					
Hydrogen	U_{H_2} [m/s]	107.0	160.0	214.0	
Reynolds Number	Re	1900	2900	3900	

Table 2
Experimental conditions for airflows

Total air flow rate Q_a [l/min]	120							
Velocity diffrence ΔU_a [m/s]	2	4	6	8	10	12	14	16
High air velocity U_H [m/s]	4.7	6.4	8.1	9.8	11.5	13.2	14.9	16.6
Low air velocity U_L [m/s]	2.7	2.4	2.1	1.8	1.5	1.2	0.9	0.6

which was performed using modified KIVA3V code [8] is based on the flamelet model under the assumption of one-step irreversible reaction. Thus, the equations of the mixture fraction and the variance were added. The estimation of the flow field is based on the two-equation model. The flamelet library was generated for the case of a global one-step irreversible reaction of propane combustion. The effect of radiation is not considered. The calculation was two-dimensionally implemented for half of the cylindrical furnace to a height of 500 mm under the assumption of axisymmetry. Only the velocity vector map is used in the present paper because the temperature was overestimated but agreed well with the flame size. This is thought to be caused by the modeling of combustion, and the result is considered to be qualitative. This numerical analysis must be improved in future studies.

4. Results and discussion

4.1. Propane flame

Figure 2 shows photographs of propane flames at an air velocity difference $\Delta U_a = 10$ m/s. The increase in D_1 leads to a contraction of the luminous flame region and an expansion of the blue flame region. This tendency is enhanced with ϕ, and with ΔU_a (not shown herein). In particular, in the case of $\phi = 0.8$ and $D_1 = 182$ mm, the luminous flame region vanishes at $\Delta U_a = 12$ m/s and 14 m/s (not shown herein). The flame size is enlarged with ϕ because of the increase in the fuel flow rate at the fixed total airflow rate. In the case of $D_1 = 95$ mm, the flame touches the wall with an increase in luminosity. On the other hand, in the case of $D_1 = 182$ mm, the increase in ϕ decreases the luminosity, but increases the flame size. The increase in ΔU_a (not shown herein) showed a decrease in the flame length, being attributable to the contraction of the luminous region.

Figure 3 shows a velocity vector map at the bottom of the furnace for the case of $D_1 = 95$ mm, $\Delta U_a = 8$ m/s, and $\phi = 0.4$. A large vortex is generated in the area between the fuel jet and the wall until 20 cm downstream from the inlet nozzles. Therefore, the above flame appearance with respect to D_1, ΔU_a, and ϕ should be attributable to the entrainment of the burned gas recirculated by the vortex, because it is well known that the decay of the luminous flame is caused by dilution. The air velocity difference ΔU_a may affect the flame appearance through the flame stretch, but also enhances the mixing. The increase in ϕ under the present experimental condition of the fixed total air flow rate decreases the oxygen concentration in the burned gas. Consequently, the combination of large D_1, ΔU_a, and ϕ enhances the effect of dilution.

Figure 4 shows the emission index of NO_x with respect to D_1, ΔU_a, and ϕ. The $EINO_x$ characteristics correspond roughly to the above flame appearance, namely the $EINO_x$ and the luminosity. The increase in D_1 clearly decreases the $EINO_x$. The effect of ϕ on the $EINO_x$ is small in the case of $D_1 = 95$ mm and is larger with a decrease in ϕ in the case of $D_1 = 182$ mm, corresponding to the flame luminosity. Therefore, the

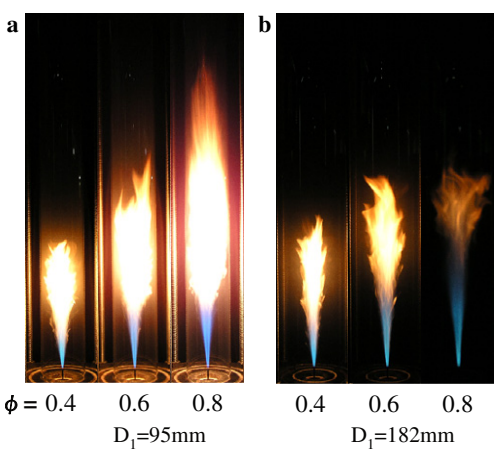

Fig. 2. Photographs of propane flames. The air velocity difference ΔU_a is 10 m/s.

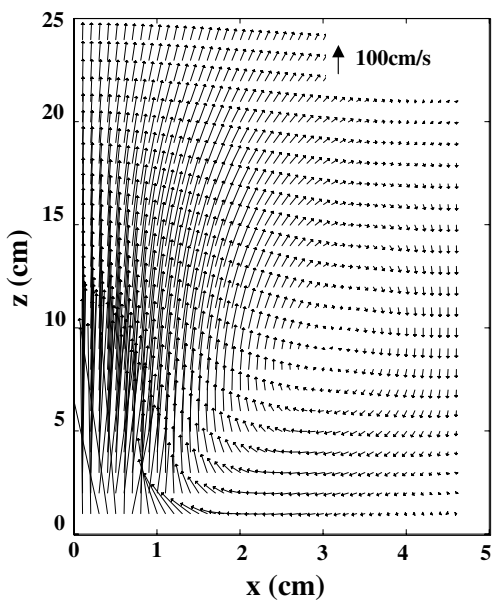

Fig. 3. Velocity vector map of a propane flame at the bottom of the furnace for $D_1 = 95$ mm, $\Delta U_a = 8$ m/s, and $\phi = 0.4$. The calculation is based on the flamelet model under the assumption of one-step irreversible reaction.

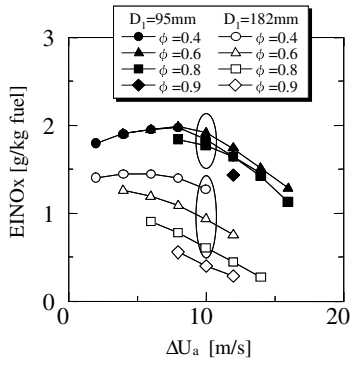

Fig. 4. $EINO_x$ characteristics of propane flames. The two circles indicate the experimental conditions under which the photographs shown in Fig. 2 were taken.

$EINO_x$ characteristics depend strongly on the dilution through the recirculation of the burned gas in the furnace. In addition, the losses with respect to the heat radiation and the heat conduction must be considered. The increase in the luminosity that is mainly attributable to soot produced in the flame enhances the radiation heat loss. The flame interaction with the present pyrex wall also enhances the heat loss owing to the conduction. Both of these effects lead to a decrease in flame temperature and thus a decrease in the $EINO_x$. In the case of $D_1 = 95$ mm, the radiation and conduction effects may be confirmed by a slight decrease in the $EINO_x$ with ϕ, but are not large. In the case of $D_1 = 182$ mm, however, the $EINO_x$ decreases with the decrease in luminosity. This also reveals that the present NO_x characteristics are dominated primarily by dilution, rather than heat losses. However, these effects must appear differently in the configuration of the adiabatic wall. This remains a subject for future investigation. On the other hand, the effect of ΔU_a is slightly complicated. In the cases of $D_1 = 95$ mm and $D_1 = 182$ mm for $\phi = 0.4$, the $EINO_x$ increases with ΔU_a in the lower range of ϕ and decreases in the higher range of ϕ. This is explained by the two factors of the flame stretch and the mixing. Lower flame stretch enhances the reaction, and higher flame stretch diminishes the reaction. The increase in the stretch eventually leads to the extinction of the flame [9]. Therefore, the activation of reaction by the lower stretch leads an increase in the $EINO_x$. The higher flame stretch causes flame cooling and dilution by the burned gas, because of the enhanced turbulent mixing.

As mentioned above, the NO_x emission is strongly related to the flame appearance and the dilution. This fact may imply that the furnace combustion is modeled by some dominant parameters. However, we failed to characterize the present furnace combustion according to the conventional parameters. Thus, we introduced two new parameters; the furnace Reynolds number Re, $c = D_1 U_F/\nu$ and the air strain rate $S = \Delta U_a/\Delta L$ ($\sim Da^{-1}$). Figures 5 and 6 show the $EINO_x$ with respect to Re, c and S, respectively. Generally, the $EINO_x$ tends to decrease with both of the new parameters, with a slight increase in the lower region of both of the parameters where the turbulence is relatively weak. The present Reynolds number reflects the turbulence intensity in the furnace and/or the intensity of the vortex formed at the bottom of the furnace, because $D_1 U_F$ in Re, c should be proportional to the circulation of the vortex. Consequently, the Reynolds number indicates primarily the dilution by the burned gas and, to a lesser degree, the flame stretch. The strain rate S reflects mainly the flame stretch, and the dilution is reflected to a lesser degree through the enhancement of the turbulent mixing. The increases in $EINO_x$ in the range of both lower Re, c and lower S are caused by the temperature increase due to the activation of reaction by the flame stretch, as mentioned above. On the other hand, in the range of both higher Re, c and higher S, both the parameters affect the $EINO_x$ with some certain tendencies to decrease. Therefore, the $EINO_x$ must be a function of Re,

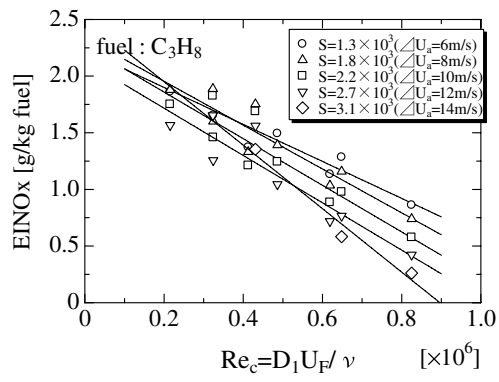

Fig. 5. $EINO_x$ with respect to Re, c for propane flames.

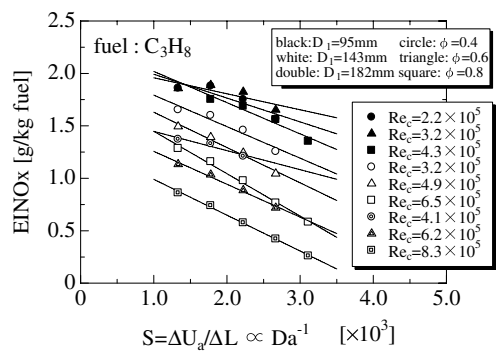

Fig. 6. $EINO_x$ with respect to strain rate S for propane flames.

c and S. The $EINO_x$ is shown in Fig. 7 as a function of $D_1 U_F \Delta U_a$, and the data are distributed along a line. This characteristic is important for understanding the flame structure of the furnace combustion. The NO_x emission is determined by the dimension of the furnace, not the flame volume, and the stretch rate. This shows that the reaction is dominated by the characteristics of the entire region of the furnace, as well as the flame stretch. Figure 8 shows the reaction rate of NO_x based on the resident time of fuel in the flame volume as a function of $D_1 U_F \Delta U_a$, according to the conventional models. The reaction rate does not show any tendency. Although this result is predicted from Fig. 7, this indicates that the reaction rate does not have a linear relationship with respect to Re, c or Da, unlike the hydrogen coflowing air jet flame [3]. These results suggest that the conventional models based on the flamelet model should be modified to take the furnace combustion into account.

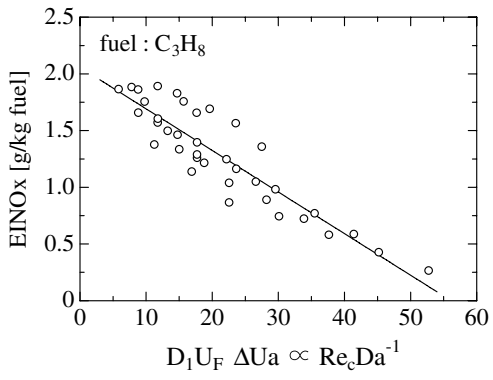

Fig. 7. $EINO_x$ with respect to $D_1 U_F \Delta U_a$ for propane flames.

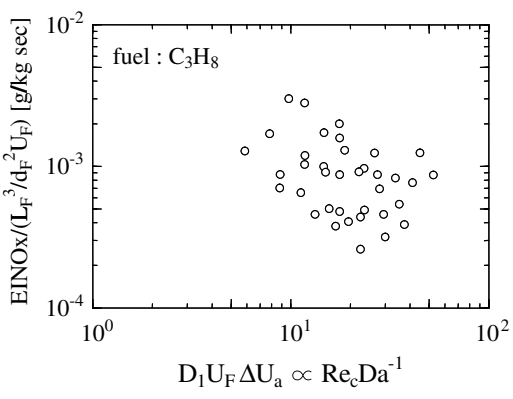

Fig. 8. Production rate of NO_x with respect to $D_1 U_F \Delta U_a$ for propane flames.

4.2. Hydrogen flame

The above results for propane flames may depend on the fuel species. The chemistry of the hydrogen flame is believed to be faster than the propane flame because the burning velocity of the hydrogen/air mixture is approximately seven times faster than of the propane/air mixture. Therefore, the characteristics of the hydrogen flame are discussed in the following.

Figure 9 shows photographs of hydrogen flames. The flames are highly turbulent and obscure at the flame boundary, differing from propane flames. This is due to high exit velocities and a high burning velocity. The flame volume is depressed compared to that of the propane flames, but increases with ϕ, in the same manner as propane flames. Not every flame touches the wall. This should facilitate the formation of vortices in the furnace. The luminosity increases with ϕ in the case of $D_1 = 95$ mm, but decreases in the case of $D_1 = 182$ mm. The increase in ΔU_a shortens the flame length and diminishes the flame color, in the same manner as propane flames (not shown herein). The effect of each parameter on the flame appearance is very similar to the effects on the propane flame. Figure 10 shows the $EINO_x$ with respect to D_1, ΔU_a, and ϕ. The $EINO_x$ decreases with ΔU_a monotonically, in contrast to the propane flame. This may be caused by the dilution enhanced by the vortex, which is easily generated and scaled up because of the smaller flame size compared to that of the propane flame. The increase in D_1 also decreases the $EINO_x$. In addition, the increase in ϕ tends to decrease the $EINO_x$ to a greater degree in the range of smaller ΔU_a. The effect of ϕ is however diminished in the case of $D_1 = 95$ mm, and disappears between $\phi = 0.4$ and 0.6. This change is not explained herein.

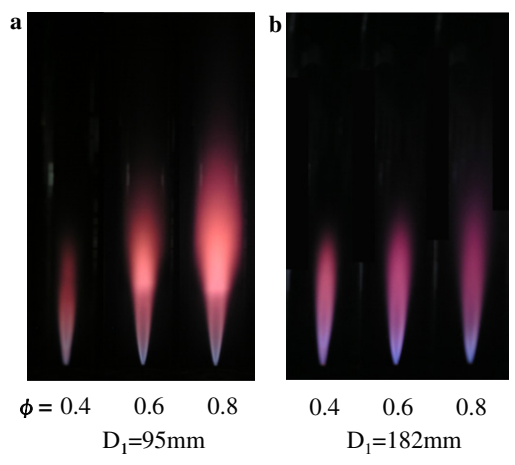

Fig. 9. Photographs of hydrogen flames. The air velocity difference ΔU_a is 2 m/s.

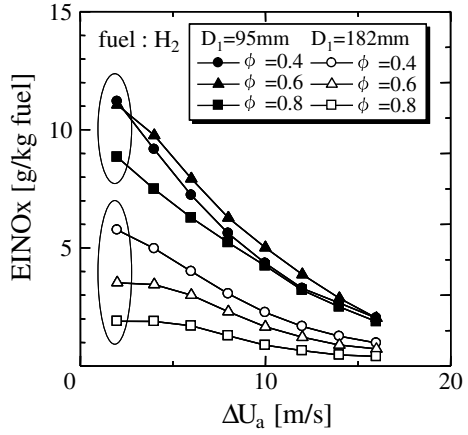

Fig. 10. $EINO_x$ characteristics of hydrogen flames. The two circles indicate the experimental conditions under which the photographs shown in Fig. 9 were taken.

Figures 11 and 12 show the $EINO_x$ with respect to *Re*, *c* and *S*, respectively, on a logarithmic scale for easy curve fitting. The $EINO_x$ decreases slowly in the lower ranges of *Re*, *c* and S and decreases rapidly in the higher range. A tendency exits between *Re*, *c* and *S*, with the exception of the data for $D_1 = 95$ mm in Fig. 12. These results again resemble those for the propane flames. These data are therefore distributed around a line with respect to the parameter $D_1 U_F \Delta U_a$, as shown in Fig. 13. This result shows that the characteristics of hydrogen flames are dominated by the furnace volume rather than the flame volume. Both the data for the propane flames and the hydrogen flames are very similar, even though the reaction rate is different. Chen and Driscoll have shown that the characteristics of the hydrogen coflowing air jet diffusion flame are predicted by the flamelet model [3]. The present results imply that the combustion in the furnace should be modified from the conventional flamelet model.

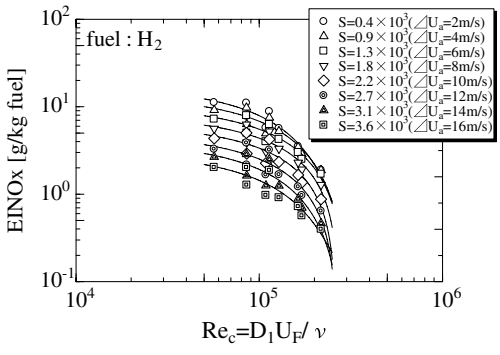

Fig. 11. $EINO_x$ with respect to *Re*, *c* for hydrogen flames.

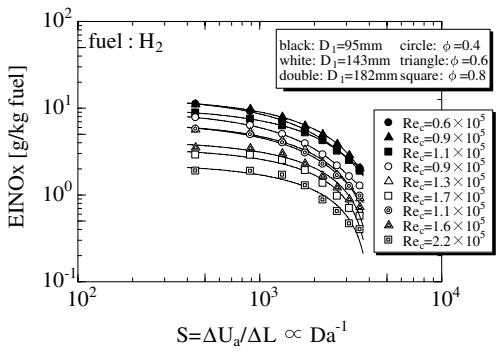

Fig. 12. $EINO_x$ with respect to strain rate *S* for hydrogen flames.

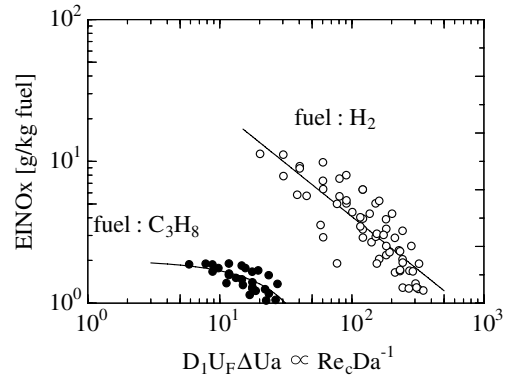

Fig. 13. $EINO_x$ with respect to $D_1 U_F \Delta U_a$ for hydrogen flames with propane data.

5. Conclusions

The flame characteristics of confined propane and hydrogen jet nonpremixed flames were investigated in cylindrical furnaces. The flame characteristics can be strongly dominated by the combination of the burner and furnace geometries. In the present study, the effects of the inner diameter of the cylindrical furnace D_1, the turbulence at the flame boundary, and the global equivalence ratio ϕ are examined in terms of emissions of NO_x. The emission index of NO_x, $EINO_x$, decreases roughly with these parameters, but the effect of the global equivalence ratio is diminished in the case of the smaller-inner-diameter furnace. The decrease in the $EINO_x$ was shown to be related to the dilution of mixtures by the burned gas and the flame stretch. The dilution is attributable to vortices formed at the bottom of the furnace, and the flame stretch is attributable to the air velocity difference ΔU_a generated by the two air nozzles. The $EINO_x$ characteristics are evaluated by the furnace Reynolds number *Re*, *c* based on

the inner diameter of the furnace, reflecting the dilution and the strain rate $S \sim Da^{-1}$ based on the air velocity difference, where Da indicates the Damköhler number. The $EINO_x$ is well characterized by the factor $D_1 U_F \Delta U_a$, which is proportional to Re, cDa^{-1}. The confined nonpremixed flame has been found to be characterized by the furnace volume, rather that the flame volume, which is related to the reaction rate of the jet nonpremixed flame with coflowing air, based on the flamelet model [2]. This may imply that the characteristics of the furnace combustion should be modified from the flamelet model due to the dilution.

Acknowledgments

The authors thank professor emeritus Y. Onuma for valuable discussions. The present study was supported by NGK Insulators, LTD.

References

[1] H. Tsuji, A. Gupta, T. Hasegawa, M. Katsuki, K. Kishimoto, M. Morita, *High Temperature Air Combustion*, CRC Press, 2003.
[2] N. Peters, S. Donnerhack, *Proc. Combust. Inst.* 18 (1981) 33–42.
[3] R.-H. Chen, J.F. Driscoll, *Proc. Combust. Inst.* 23 (1990) 281–288.
[4] S.R. Turns, *Prog. Energy Combust. Sci.* 21 (1995) 361–385.
[5] S.H. Kim, Y. Yoon, I.-S. Jeung, *Proc. Combust. Inst.* 28 (2000) 463–471.
[6] A. Santos, M. Costa, *Combust. Flame* 142 (2005) 160–169.
[7] T.X. Phuoc, M.P. Mathur, J.M. Ekmann, F.P. White, *Optical Diag. Eng.* 2 (1) (1997) 32–43.
[8] A.A. Amsden, *KIVA-3*, Los Alamos National laboratory, 1998.
[9] T. Takagi, Y. Yoshikawa, K. Yoshida, K. Komiyama, S. Kinoshita, *Proc. Combust. Inst.* 26 (1996) 1103–1110.